Issue	**1**
Factory ID	**25298**
Date of Issue	**10-Jun-2026**

Smart Apparel (Pvt.) Ltd.

Pakistan ACCORD for Health and Safety in the Garment and Textile Industry

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Faisalabad
Punjab Pakistan

Latitude: [31.50359], longitude: [73.119422]
Date of inspection: 14-Apr-2026

Electrical Safety Inspection Report

Observations & Actions



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Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 14-04-2026 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

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1. Executive Summary

On 14-Apr-2026, Azka Riaz, Muhammad Bilal Butt and Hamza Talib on behalf of the Pakistan Accord carried out an initial electrical inspection for Smart Apparel (Pvt.) Ltd., a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with factory management representative Mr. Muhammad Shoaib Ijaz Compliance consultant, Mr. Bilal Yaqoob Health and Safety, Mr. Nazar Ali Electrical, Mr. Muhammad Muaz Compliance Officer along with other staff members of the factory. As the assessment was being carried out as part of an overall Structural, Fire and Electrical initial inspection, other representatives from Pakistan Accord were also present and the team was led by Tariq Mehmood Pakistan Accord.

The factory complex is entirely occupied by Smart Apparels Pvt. Ltd. Enterprises as a tenant.

The factory operates with single shift of 8 hours for all the activities. Approximately 90 workers operate in the factory out of which approximately 18 employees are female.

The factory comprises three main buildings and four ancillary buildings, details of which are provided below:

1. **Building 1** is a two-storey G+1 building, used for Sampling, Offices, Warehouse, Metal free zone on ground level, stitching on first floor, and the roof is inaccessible.
2. **Building 2** is a two-storey G+1 building with steel portal frame shed on first floor. The building is used for main store, stitching, sampling, offices on ground floor, cutting and checking on first floor, and the roof is inaccessible.
3. **Building 3** is a ground-only building, used for pressing, checking, packing, warehouse, carton storage and offices, and the roof is unoccupied.
4. **Ancillaries**

a. **Office,** ground only structure, used as office.
b. **Washroom,** ground only structure, used as washroom and water tank.
c. **Utilities,** ground only steel shed, used for boiler, compressor and RO plant.
d. **Mechanical Plant,** ground only structure, used as washrooms and mechanical room.

A summary of the main concerns identified during the inspection includes (non-exhaustive):

- **Fire Alarm & Detection System (FADS):** A conventional fire alarm panel is installed in the facility; however, it was found non-operational during the visit. Several deficiencies were observed, including issues with panels, cabling, and hooters, along with the use of non-fire-rated cables. However, the 220V hooter system was found operational. These conditions may affect the performance and reliability of the system during a fire emergency.
- **Hot Spots:** Observed in electrical panels and distribution boards during the thermographic testing. These could be the result of several issues such as loose connections, overloaded circuits, aging components, imbalanced loads, faulty part or poor ventilation of panels that may lead to equipment failure and fire hazards.

- **Insufficient emergency and directional signage:** During the assessment, it was observed that emergency and directional signage, along with emergency lighting, was insufficient in certain areas. This may hinder safe evacuation during power outages and increase the risk of accidents.
- **Earthing System:** Distribution boards and equipment were lacking proper earthing which increases the risk of electrical shocks, equipment damage, and fire hazards. Immediate grounding and proper earthing are essential for safety and compliance with electrical standards.

We see no reason to suspend operations in the facility due to these concerns (subject to the required actions noted in this report being addressed within agreed timeframes.)

Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents

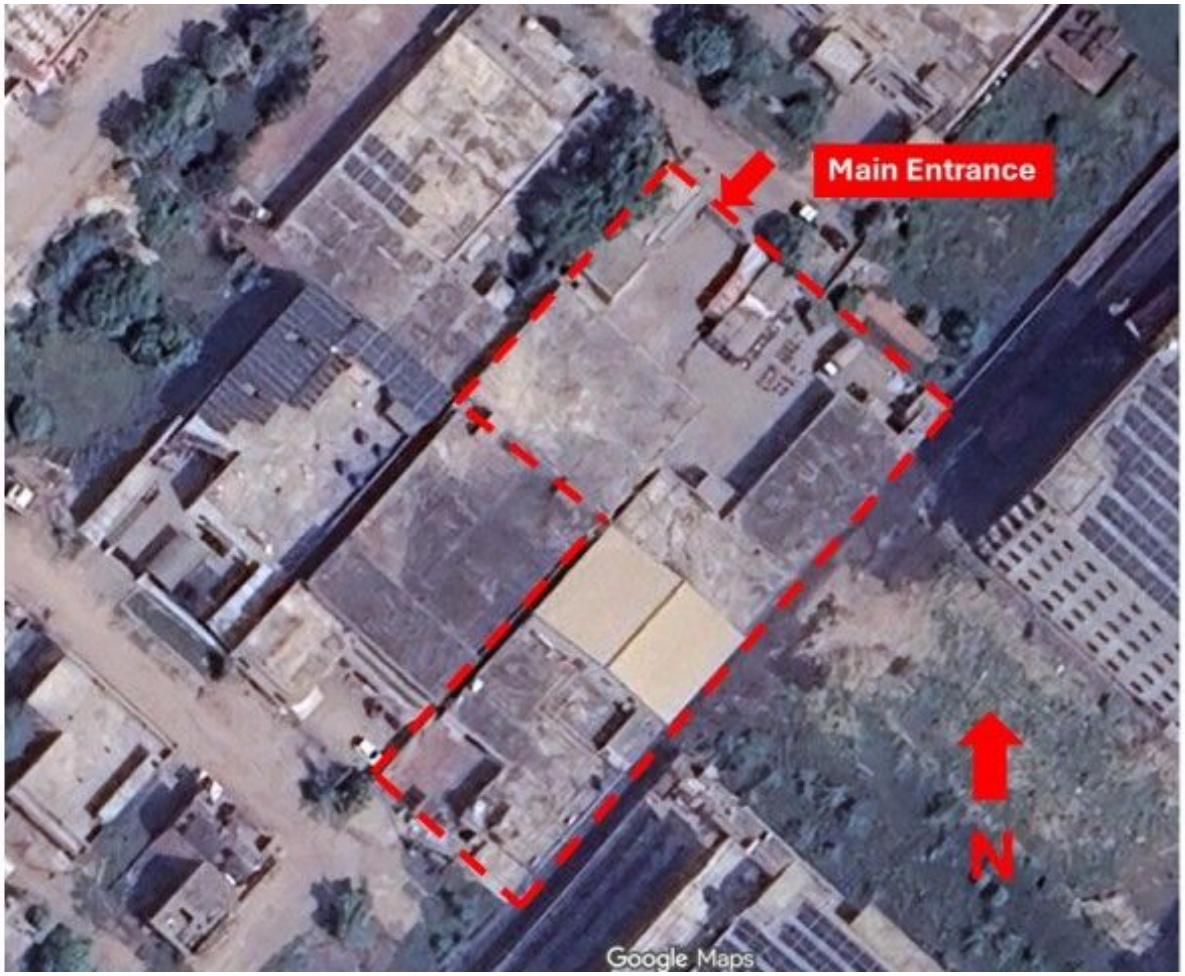


Figure 1.1. Site layout (satellite view) with red dashed lies indicates approximate site boundary

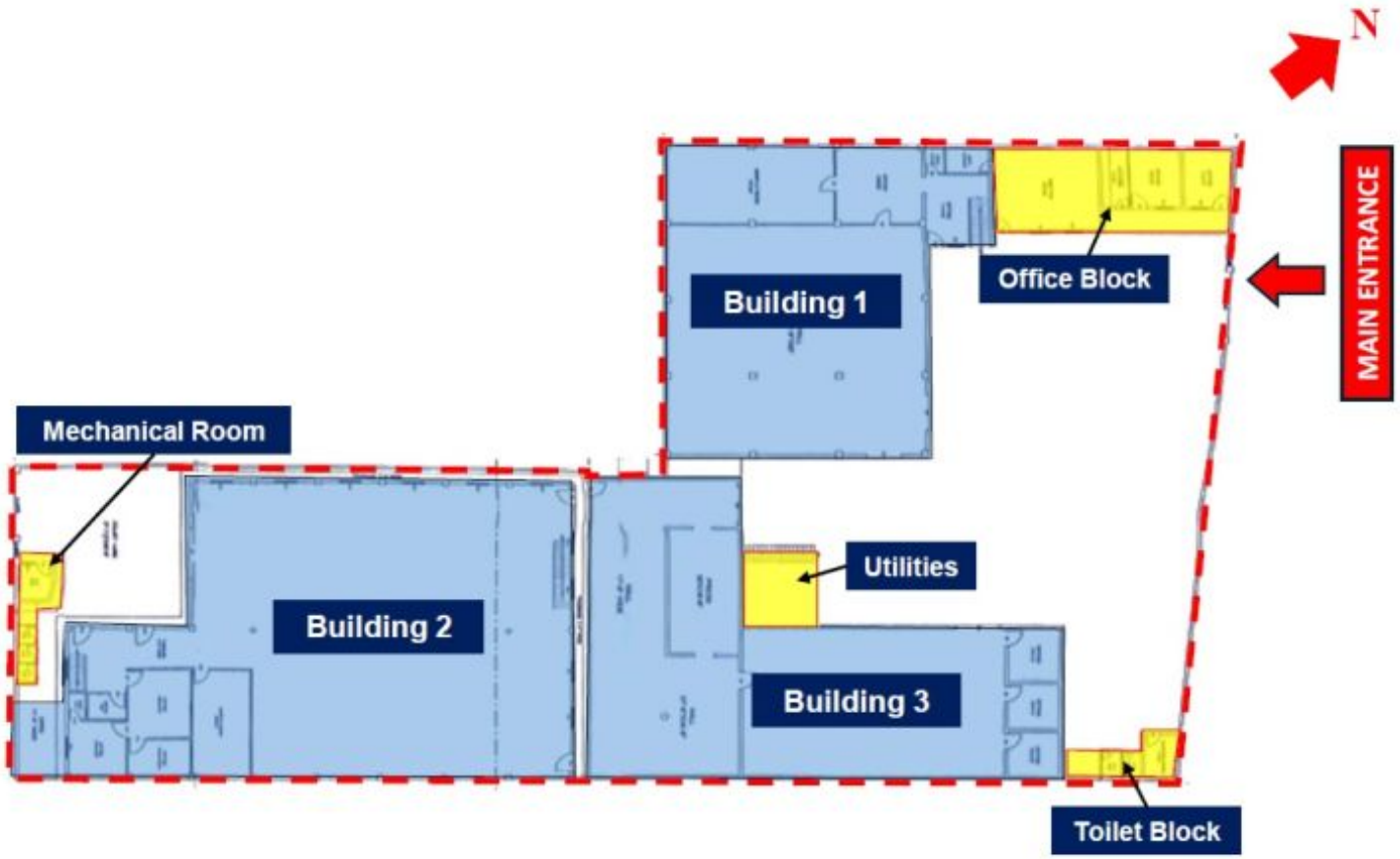


Figure 1.2. Site layout plan - Red dashed line indicates site boundary, highlighted area indicates buildings inspected

2.1 Building Use

The factory comprises three main buildings and four ancillary buildings, details of which are provided below:

Building 1
L00- Ground floor- sampling, offices, warehouse, and metal-free zone
L01- First floor- stitching
L99- Roof- inaccessible

Building 2
L00- Ground floor- main store, stitching, sampling, and offices
L01- First floor- cutting and checking
L99- Roof- inaccessible

Building 3
L00- Ground floor- pressing, checking, packing, warehouse, carton storage, and offices
L99- Roof- unoccupied

Ancillary Building
L00- Ground floor- office

Ancillary Building
L00- Ground floor- washroom and water tank

Ancillary Building
L00- Ground floor- boiler, compressor, and RO plant

Ancillary Building
L00- Ground floor- washrooms and mechanical room

2.2 Utility Source

This facility has two FESCO supplies at 400/440V serving as the primary power sources. One connection is rated at 100 kVA and is connected to the main LT panel located in the electrical room. The second FESCO connection, also rated at 100 kVA, is connected to a main distribution board installed in the open area. Both supplies are operated alternately, with each source being utilized for approximately 15 days per month to support the facility's electrical load.

2.3 Backup source

This facility has one standby diesel generator of 75 kVA as backup power. The generator operates in the event of a mains power failure. The fire alarm system is supported by 220V hooters, and the emergency lighting system is provided with battery backup supply.

3. Documentation

This table shows the details of the documentation provided by the factory.

S. No	Description	Availability Yes/No	Created/Updated Date	Status/Remarks
1	As-Built Layout Drawings	No	-	-
2	Single Line Diagrams	No	-	-
3	Fire Alarm Drawings	No	-	-
4	Earth Test Report	No	-	-
5	Thermographic Test Report	No	-	-
6	Insulation Resistance Test Report	No	-	-
7	LPS Risk Assessments	No	-	-
8	Periodic Maintenance Record	No	-	-
9	Transformer Oil Testing Report	N/A	-	-

4. Electrical Safety Measures

During our electrical site inspection of Smart Apparel (Pvt.) Ltd., a number of electrical issues were identified.

It is essential to address these electrical safety issues promptly to enhance safety and compliance within Smart Apparel (Pvt.) Ltd.. Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

It is essential to address these electrical issues promptly to enhance safety and compliance within the Smart Apparel (Pvt.) Ltd. factory. Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions.

All survey, testing, and design activities specified in this report shall be performed by a qualified electrical engineer. The recommended installation and modification tasks should be executed by a trained professional electrician.

Before commencing any installation and/or modification works, the proposed design must be submitted to the Accord for approval.

Upon completion of the works, the necessary evidence required to verify that the tasks have been completed to the agreed standards will be determined during the development of the Corrective Action Plan (CAP).

5. Remediation Support

5.1 Engineering Support

Some items in the Corrective Action Plan (CAP) will require remediation by a qualified engineer. Please refer to the Remediation Plan for guidance on which tasks can be completed using internal resources and which require engineering support. F-202604-065378

Timeline	Remediation Plan
Medium Term CAP-0077348	Factories should begin remediation of CAP items—such as cleaning, cable organization, and basic corrections—using internal resources. For systems or design-related remediation, engagement of a qualified Electrical Design Engineer is required.

6. Diagrams & Drawings

6.1 Sub-station room layout drg

This section refers to the electrical rooms within the facility.

During the inspection conducted, it was observed that there were no substation room layout drawings available for the factory.

There are no visible indications of substation equipment locations or pathways for maintenance and emergency access.

The absence of substation room layout drawings poses significant risks to life and fire safety within the factory premises. Without these critical documents, emergency responders and maintenance personnel may face difficulties in locating and isolating electrical faults during emergencies. This delay can result in prolonged exposure to electrical hazards, increasing the risk of electrical fires and endangering lives. Furthermore, the lack of layout drawings complicates routine maintenance, potentially leading to undetected faults and increased likelihood of equipment failure. F-202604-065331

Timeline	Remediation Plan
Medium Term CAP-0077349	Develop detailed substation room layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). Ensure that all critical equipment and pathways are accurately documented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0077350	Regularly update and review the substation room layout drawings to reflect any changes in equipment or configuration. Provide training sessions for maintenance personnel and emergency responders on how to interpret and utilize the layout drawings.

6.2 Single Line Diagram (SLD)

This section refers to the entire facility.

It was found that the electrical single line diagrams (SLDs) for the factory were missing.

This deficiency was noted in all electrical rooms and areas where SLDs should be prominently displayed.

The factory lacks documented diagrams that illustrate the electrical distribution system, including the connection and configuration of all electrical components.

Single line diagrams are crucial for understanding the electrical distribution network and for troubleshooting issues effectively. Without these diagrams, maintenance personnel and emergency responders lack critical information necessary for isolating faults and managing power during emergencies. This can lead to delays in addressing electrical issues, increasing the risk of electrical fires and potential harm to personnel. Moreover, the lack of SLDs complicates routine maintenance and inspection processes, potentially resulting in undetected electrical faults and system inefficiencies.F-202604-065332

Timeline	Remediation Plan
Medium Term CAP-0077351	Develop comprehensive electrical single line diagrams for the entire facility in accordance with the requirements outlined in Section 514.9.1 of BS7671 18th Edition + A2 (2022). Ensure all components and connections are accurately represented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0077352	Implement a regular review and update process for the single line diagrams to reflect any changes in the electrical system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the diagrams.

6.3 Electrical layout drawing

This section refers to the entire facility.

During the survey, it was noted that the factory lacked electrical layout drawings - including small power layouts, lighting layouts, and containment layouts.

This absence was consistent across all areas where such documentation should be available.

These drawings are vital for understanding the placement and connections of electrical systems, which is crucial for identifying and mitigating potential hazards. In emergencies, the lack of detailed layouts can delay the identification and isolation of electrical faults, increasing the risk of fire and electrical accidents. Additionally, without proper layout drawings, routine inspections and maintenance may overlook critical issues, potentially leading to system failures and safety breaches.F-202604-065340

Timeline	Remediation Plan
Medium Term CAP-0077353	Develop comprehensive electrical layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). To avoid drawings becoming overcrowded, separate drawings shall be prepared for each electrical system. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0077354	Implement a regular review and update process for the electrical layout drawings to reflect any changes in the system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the layouts.

7. DBs & Elec Protection Systems

7.1 Circuit breakers

This section refers to the entire facility.

During the survey the circuit breakers installed throughout the facility inside the electrical panels and distribution boards (DBs) were inspected.

The following observations were noted:
i. It was observed that the circuit breaker is improperly mounted without secure fixing at changeover panel near mechanical workshop. This condition may lead to loose connections, overheating, arcing, short circuits, and increased risk of electric shock due to exposure of live parts and mechanical stress on terminations.

ii. It was found that a single-phase load was fed from a three-pole breaker. Using a single phase of a three-pole breaker can lead to unbalanced load conditions. This can cause the breaker to trip unexpectedly or, in some cases, fail to trip when necessary, compromising the safety of the system. The issue was noted at first floor stitching area and ground floor finishing area distribution boards.

iii. It was observed that multiple circuit breakers had ratings that were not appropriate for the connected cable sizes. For example,
a) In the LT panel, a 50A circuit breaker was connected to 4mm² cables.
b) In the finishing area distribution board, a 100A circuit breaker was connected to 10mm² cables.
c) In the cutting area distribution board, a 100A circuit breaker was connected to 4mm² cables.

Specifically, some circuit breakers had higher ratings than the cables could safely handle, which poses a significant risk of overheating and potential fire hazards. Conversely, other circuit breakers had lower ratings than necessary, resulting in frequent and unnecessary tripping. To ensure both safety and operational efficiency, corrective action is required to align the circuit breaker ratings with the cable capacities. This adjustment will mitigate the risks associated with overheating and improve the reliability of the electrical system. F-202604-065327

Timeline	Remediation Plan
Medium Term CAP-0077586	i. Securely mount the circuit breaker in a proper enclosure with adequate IP rating, ensuring all cable entries are gland-sealed, conduits are properly terminated and supported, and no conductors remain exposed.

Medium Term CAP-0077587	ii. Reconfigure the circuit to supply single-phase loads through appropriately rated single- or double-pole breakers instead of using a three-pole breaker. Ensure proper phase load balancing and verify protection coordination in accordance with good electrical engineering practices.
Medium Term CAP-0077588	iii. Evaluate the breaker ratings and cable sizes for all circuits in the facility according to Section 433.1.1 and Appendix 4 of BS 7671 18th Edition + A2 (2022). Ensure that the breaker rating does not exceed the cable's current-carrying capacity. If the breaker has adjustable settings, they must be configured to allow current within the cable's ampacity after applying derating factors.



Figure 2.1 - A 50A circuit breaker was connected to 4mm² cables at LT panel



Figure 2.2 - A 100A circuit breaker was connected to 10mm² cable at the finishing area distribution board



Figure 2.3 - Circuit breaker installed without proper mechanical support inside electrical changeover panel



Figure 2.4 - 100A circuit breaker was connected to 4mm² cables at cutting area distribution board



Figure 2.5 - Single-phase load connected through a three-pole circuit breaker at finishing area ground floor

7.2 Hot Spots

This relates to the entire facility.

During the inspection, Thermographic surveys were conducted on the electrical boards.

The survey identified numerous hot spots within panels and distribution boards. Hot spots in electrical distribution systems can be caused by several factors, including:

> Loose connections
> Overloaded circuits
> Faulty components
> Improper sized cable
> Insulation degradation

It was also found that the thermographic report was not available at the site. Without thermography testing, faults such as overheating components or loose connections may go unnoticed, which increases the risk of fire. Early detection through thermography can prevent fire incidents.

The categories of identified hot spots have been evaluated as per the ANSI NETA MTS 2019 standard, Table 100.18. Hot spots were observed, particularly at the following locations:
a) 129°C at the cutting area first floor distribution board
b) 122°C at the RO plant distribution board
c) 88.3°C at the cutting area first floor distribution board
d) 65.7°C at the changeover panel near the mechanical room

Circuit breakers are typically designed to operate safely within an ambient temperature range of -20°C to 40°C. Commonly used PVC insulated cables typically have a maximum operating temperature of 70°C. Prolonged exposure to temperatures above these can degrade the insulation of the breaker as well as the cable and result in reduced performance.

Hot spots in electrical systems pose significant life safety risks due to several factors. Overheated components can ignite surrounding materials, leading to electrical fires. This is particularly dangerous in densely packed panels and distribution boards where flammable materials may be present. Hot spots can degrade insulation on wires and components, increasing the risk of electrical shock to personnel who might come into contact with exposed conductive parts. Overheating can contribute to conditions that might trigger an arc flash, a dangerous high-temperature

event that can cause severe injuries or fatalities to personnel working near the affected equipment.F-202604-065339

Timeline	Remediation Plan
Immediate CAP-0077604	Any circuits identified on site with temperatures exceeding 40°C above the ambient temperature, operating above 90°C or exhibiting visual signs of burning must be immediately isolated and taken out of service. These circuits should remain de-energized until a thorough review has been conducted and appropriate remedial actions have been implemented to resolve the underlying issues.
Medium Term CAP-0077605	During the inspection, several circuits were identified where the temperature was exceeding 20°C above ambient temperature. It is recommended that the factory conducts comprehensive thermographic surveys of all electrical panels to detect similar issues. Identified hot spots should be thoroughly reviewed to determine their underlying causes. Corrective actions must be taken to address and resolve these issues, ensuring safe and reliable operation of the electrical systems.
Longer Term CAP-0077606	Regular thermographic inspections should be incorporated into the maintenance schedule to proactively manage and mitigate the risks associated with overheating. Submit the thermographic test report to the Pakistan Accord for review.



Figure 2.6 - Hot spot found on neutral busbar at cutting area first floor distribution board (thermal photo)



Figure 2.7 - Hot spot found on neutral busbar at cutting area first floor distribution board (digital photo)



Figure 2.8 - Hot spot found on termination at RO plant distribution board (thermal photo)



Figure 2.9 - Hot spot found on termination at RO plant distribution board (digital photo)



Figure 2.10 - Hot spot found on cable at cutting area first floor distribution board (thermal photo)



Figure 2.11 - Hot spot found on cable at cutting area first floor distribution board (digital photo)

7.3 Earthing system

This relates to the entire facility.

During the inspection of the facility's electrical panels, distribution boards, and machinery, several observations with the earthing system were identified.

i. Comprehensive drawings for the earthing system were absent. The lack of proper documentation and identification of the earthing and neutral system components pose safety risks.
ii. The earth pits were inaccessible and unsuitably located outside building 2.
iii. The earthing cables were not appropriately sized according to the relevant earthing standards. The issue was noted at LT panel.
iv. The earthing system was absent in multiple locations. Several distribution boards lacked earthing cables, and many electrical machines, power sockets did not have proper earth connections. The issue was noted at finishing area, stitching area, main changeover panel near mechanical workshop and cutting area distribution boards, machines and diesel generator.
v. It was noted that the earth bar was incorrectly connected to the neutral bar, indicating improper grounding practices. The issue was noted at cutting area distribution board and main changeover panel near mechanical workshop.
vi. Equipotential bonding for the cable trays and metallic panel covers was missing at LT panel, stitching area, finishing area, main changeover panel and cutting area.
vii. It was found that the earth testing record was not available at the site. Without earth testing records, it is difficult to ensure that the grounding system is functioning correctly and that maintenance activities.

The lack of proper earthing system throughout the facility poses substantial safety risks, such as electric shocks, fires, and equipment damage. Implementing appropriate earthing measures is crucial to prevent hazardous voltage levels and ensure operational safety. F-202604-065341

Timeline	Remediation Plan
Medium Term CAP-0077619	i. Design and evaluate the earthing system installation throughout the facility by preparing a comprehensive earthing system diagram, including identification of all earthing and neutral system components. Ensure compliance with BS 7671: 18th Edition + A2 (2022), Regulation 514.5.1 regarding proper identification and labeling and, submit it to Pakistan Accord for review.
Medium Term CAP-0077620	ii. Relocate and redesign earth pits to make them easily accessible for maintenance and inspection, complying with BS 7671 18th Edition + A2 (2022) Chapter 54.

Medium Term CAP-0077621	iii. Replace all earthing cables with appropriately sized cables as per the requirements of BS 7671 18th Edition + A2 (2022) Appendix 4, Section 5. Verify that all earthing conductors meet the minimum cross-sectional area specified.
Medium Term CAP-0077622	iv. Install earthing systems where absent, ensuring compliance with BS 7671 18th Edition + A2 (2022) Chapter 54. Fit all electrical boards with the required earthing cables. Ensure that earthing conductors are properly connected and comply with BS 7671 18th Edition + A2 (2022) Section 543.1.
Medium Term CAP-0077623	v. Separate the earth bar from the neutral bar in all distribution boards in accordance with BS 7671: 18th Edition + A2 (2022).
Medium Term CAP-0077624	vi. Provide equipotential bonding for all cable trays, metallic panel covers, and other exposed conductive parts, complying with BS 7671: 18th Edition + A2 (2022)
Medium Term CAP-0077625	vii. Prepare and submit the earth testing record to the Pakistan Accord for review.
Longer Term CAP-0077626	Establish a schedule for regular maintenance and inspection of the earthing system and protective devices as per BS 7671 Regulation 651.2. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on proper earthing practices and the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.12 - Earth pit not constructed in accordance with relevant standards - outside building 2



Figure 2.14 - Earth bond missing on panel door



Figure 2.13 - Undersized earth cable for LT panel at LT panel area



Figure 2.15 - Earthing system found to be absent at changeover panel LT room



Figure 2.16 - Stitching machine earth connection and cable tray earth bonding was found missing at ground floor



Figure 2.18 - Disconnected earth cable observed in office area split AC



Figure 2.17 - Neutral found to be combined with earth at cutting area distribution board



Figure 2.19 - Disconnected earth cable observed at power sockets installed at stitching area



Figure 2.20 - Disconnected earth cable observed at diesel generator



Figure 2.21 - Earthing system absent at finishing area first floor distribution board

7.4 Panel/cabling/busbar

This relates to the entire facility.

The visual site inspection included an examination of the electrical panels, cables and busbars throughout the facility.

i. Several electrical panels exhibited signs of corrosion, indicating exposure to moisture, weather and atmosphere. The issue was noted at stitching area, cutting area, changeover panel and diesel generator panel near mechanical workshop area.

ii. Visible signs of damage were observed to electrical components, including damaged circuit breakers, signs of burning, and deterioration of cable insulation. Faulty or non-functional phase indication lights were also noted. This condition was observed at the machine power cable in the stitching hall, ground floor. The issue was noted at stitching area DB, stitching machine power cables, generator panel and changeover panel near mechanical workshop.

iii. Improper panel internal arrangement was observed, including missing phase separators and inadequate cable dressing, increasing the risk of phase contact, loose connections, and overheating. The issue was noted at changeover panel near mechanical workshop, stitching area, cutting area and finishing area distribution boards.

The utilisation of electrical equipment that is either unsuitable for its operating environment or physically damaged poses significant safety risks. These risks include potential electrical failures, increased likelihood of short circuits, and the heightened risk of fire and electrical shock. F-202604-065342

Timeline	Remediation Plan
Medium Term CAP-0077651	i. Replace or refurbish corroded electrical panels to prevent further degradation and potential safety hazards. Implement protective measures such as installing weatherproof enclosures or using corrosion-resistant materials.
Medium Term CAP-0077652	ii. Inspect all affected electrical components and replace or repair damaged circuit breakers, burnt or deteriorated cables, and faulty phase indication lights. Ensure all repairs are carried out in accordance with BS 7671 (18th Edition + A2:2022).
Medium Term CAP-0077653	iii. Ensure all distribution boards are properly dressed and organized to prevent mechanical stress on terminals. Additionally, install missing phase separators between incoming and outgoing terminations to provide adequate insulation and prevent flashovers between phases, in accordance with BS 7671:2018+A2:2022.

Longer Term CAP-0077654	Establish a schedule for regular maintenance and inspection of the panel, cabling and busbars. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.22 - Damaged power cable for stitching machine observed at stitching hall ground floor



Figure 2.23 - Faulty ampere meter at changeover panel near mechanical workshop



Figure 2.24 - Inadequate cable dressing and missing phase separators observed at changeover panel near mechanical workshop



Figure 2.26 - Inadequate cable dressing observed at cutting area distribution board



Figure 2.25 - Corrosion observed at cutting area electrical distribution board



Figure 2.27 - Corrosion observed on diesel generator panel

7.5 Loop Connection

This relates to the entire facility.

During the inspection loop connections were identified within the electrical panels.

In these loop connections, one circuit breaker is energised by the live part of another circuit breaker, creating loops within the panel instead of using a busbar system. The issue was noted at ground floor finishing area, stitching hall , and cutting area distribution boards.

Avoiding loop connections is crucial due to several significant risks. Loop connections can complicate maintenance activities, making it difficult to trace and isolate circuits. These connections can lead to increased resistance and overheating, potentially damaging components. Loop connections raise the risk of accidental phase mixing, which can result in short circuits. The above issues compromise personal safety and significantly increase the risk of fire.F-202604-065330

Timeline	Remediation Plan
Medium Term CAP-0077584	All electrical panels shall be modified to eliminate loop connections. A busbar system within the electrical panels shall be utilized to provide each circuit with the required live, neutral, and earth connections.



Figure 2.28 - Improper/unsafe loop connection observed at finishing area, ground floor



Figure 2.29 - Improper/unsafe loop connection observed at stitching hall fan distribution board



Figure 2.30 - Improper/unsafe loop connection observed at cutting area distribution board

7.6 Multiple Connections

This relates to the whole facility.

During our inspection, multilooping was observed in a number of electrical panels across the facility.

In these cases, multiple circuits were being protected by a single circuit breaker. The issue was noted at stitching hall, cutting area and ground floor finishing area distribution boards.

Multilooping can result in improper distribution of electrical current, leading to voltage drops and potential electrical hazards. To mitigate these risks and ensure the safety and reliability of the electrical system, it is essential to adhere to proper installation practices. This includes using appropriate dedicated circuit breakers for each circuit.F-202604-065328

Timeline	Remediation Plan
Medium Term CAP-0077585	The electrical installation throughout the facility shall be modified to ensure each circuit is protected by a dedicated circuit breaker of a suitable type and rating for each individual circuit in accordance with Section 430 of BS 7671 18th Edition + A2 (2022)



Figure 2.31 - Improper/unsafe loop connection observed at stitching hall fan distribution board



Figure 2.32 - Improper/unsafe loop connection observed at cutting area distribution board



Figure 2.33 - Improper/unsafe loop connection observed at finishing area, ground floor

7.7 Openings

This relates to the entire facility of the factory.

During the visual inspection, the electrical panels and distribution boards were examined for unsealed openings to ensure compliance with the IP rating requirements.

Unsealed openings were discovered in the panels and distribution boards at LT panel and finishing area.

These deficiencies pose multiple hazards. Unsealed openings and missing panels expose live electrical components, significantly increasing the risk of accidental contact and electric shock to personnel working nearby. The unsealed openings allow dust, debris, and moisture to enter, which can lead to short circuits and increase the risk of fire. The absence of proper sealing compromises the IP rating of the panels, which is crucial for ensuring safety and protection against environmental factors.F-202604-065344

Timeline	Remediation Plan
Medium Term CAP-0077642	Corrective actions are required to seal all openings and replace missing panels to restore the integrity and safety of the electrical systems. The electrical system shall ensure compliance with Sections 614.2.1, 412.2.2, and 512.2 of BS 7671 18th Edition + A2 (2022).
Longer Term CAP-0077643	Conduct regular maintenance and maintain comprehensive records for all electrical panels to ensure there are no openings and that the IP ratings of the panels remain uncompromised.



Figure 2.34 - Openings found in LT panel



Figure 2.35 - Openings found in finishing area DB first floor

7.8 Lint & dust inside panel board

This relates to the entire facility.

During the inspection, the cleanliness of electrical panels, distribution boards and electrical containment was assessed.

The inspection revealed a significant accumulation of lint and dust in most of the distribution boards, panels and trunking systems.

The presence of lint and dust inside distribution boards poses several hazards. Accumulated lint and dust can act as fuel for fires if exposed to electrical currents or sparks, significantly increasing the risk of electrical fires. The cooling capacity of electrical components can also be impaired by the presence of dust and lint, leading to overheating. The issue was noted at following locations:
a) LT panel
b) Finishing area cable tray
c) Cutting area distribution board
d) Stitching hall, first floor (cable tray)
e) stitching area distribution board

Additionally, the accumulation of dust and lint can obstruct proper operation and maintenance, thereby impairing the functionality of the electrical components.F-202604-065334

Timeline	Remediation Plan
Medium Term CAP-0077589	All electrical panels and trunking runs shall be inspected and cleaned of any dust or lint in accordance with Section 522.4.2 of BS7671 18th Edition + A2 (2022)
Longer Term CAP-0077590	A regular maintenance procedure shall be established to ensure routine inspection and cleaning of all electrical panels and trunking runs. This will prevent the accumulation of lint and dust, thereby maintaining the safety and efficiency of the electrical systems.



Figure 2.36 - Lint and dust accumulation observed at LT panel



Figure 2.37 - Lint and dust accumulation observed on cable trays at finishing area



Figure 2.38 - Lint and dust accumulation observed at cutting area



Figure 2.39 - Lint and dust accumulation observed on cable trays at stitching area

7.9 Lightning Protection System (LPS)

During the inspection, the lightning protection strategy of the factory was assessed.

No lightning protection system was installed, and there was no risk assessment provided to confirm whether it was not required.

Lightning strikes can cause significant damage to buildings, primarily through the ignition of fires. When lightning strikes a structure, it can cause an electrical surge that ignites flammable materials. Without a proper lightning protection system (LPS), the energy from a lightning strike can travel through the building's electrical and plumbing systems, creating sparks and starting fires. F-202604-065345

Timeline	Remediation Plan
Medium Term CAP-0077640	Engage a qualified lightning protection specialist to perform a comprehensive risk assessment as per BS EN 62305 (Protection against Lightning). This assessment will determine the necessity of a lightning protection system (LPS) based on factors such as building height, location, occupancy, and the presence of sensitive electronic equipment.
Longer Term CAP-0077641	If the risk assessment concludes that an LPS is necessary, proceed with the design and installation of an appropriate system in compliance with relevant standards such as BS EN 62305 (Protection against Lightning).



Figure 2.40 - No Lightning Protection System (LPS) was observed on the rooftop during inspection

8. Fire Alarm & Detection System (FADS)

8.1 Non-functional or incomplete

This section refers to the entire facility.

During the inspection, the Fire Alarm and Detection System (FADS) for the facility was evaluated.

A conventional fire alarm system comprising 220 V hooters and standalone smoke detectors is installed serving the main production buildings. During the inspection, the fire alarm control panel was found to be non-operational. Several deficiencies were also observed, including the use of non-addressable equipment, inadequate detection coverage, missing manual call points, insufficient visual notification coverage in the weaving area, and fault indications on the panel. These deficiencies collectively pose a significant life-safety risk in the event of a fire emergency.

A non-functional/incomplete FADS significantly increases the risk of worker injury or fatality, substantial property damage, operational disruptions, legal liabilities, reputational damage, and environmental harm. A delayed response during a fire event can significantly increase the impact on personnel, goods, and property.F-202604-065377

Timeline	Remediation Plan
Immediate CAP-0078642	Ensure the functionality and performance of the existing alarm and detection system within the building.
Medium Term CAP-0077510	i. Submit design documentation for the new FADS system to Pakistan Accord for review prior to procurement of any new equipment. Design documentation to include drawings, technical data sheets, and product certifications of all listed equipment to show full compliance with the Fire Alarm and Detection System (FADS) with NFPA 72.
Longer Term CAP-0077511	ii. Submit documentation to the Pakistan Accord showing that an L/C has been opened for import of any components required for the FADS design submitted earlier.
Longer Term CAP-0077512	iii. Submit documentation to the Pakistan Accord demonstrating that all materials, both imported and locally procured, have been purchased and are available on-site for installation.

Longer Term CAP-0077513	iv. Once installation of the FADS system is complete, the factory is to submit full "as-built" documentation of the system to the Pakistan Accord for review.
Longer Term CAP-0077514	v. Factory to advise Pakistan Accord when FADS installation is fully completed and ready for review. Pakistan Accord will then arrange a verification inspection to ensure that the installation fully complies with the previously reviewed design documentation.
Longer Term CAP-0077515	vi. Provide full testing and commissioning (T&C) certification from the FADS system installer and make provision for attendance of the system installer, factory technical personnel, and Pakistan Accord technical team to witness a final T&C demonstration verification on site. On successful completion of this T&C verification inspection, remediation of the FADS system will be complete.



Figure 2.41 - No smoke detection system in electrical room ground floor



Figure 2.42 - Conventional fire alarm panel installed at gate office - found to be non-operational during inspection



Figure 2.43 - Stand alone smoke detector found to be installed at packing area ground floor



Figure 2.44 - 220V hooter system installed at first floor stitching area exit

9. Emergency Lighting

9.1 Provided but non-functional

The section refers to the entire facility.

During the inspection it was observed that emergency lighting was provided but not functional at warehouse and passage between finishing and stitching areas ground floor.

Functional emergency lighting is critical for providing illumination in the event of a power failure or other emergencies, ensuring that workers can safely evacuate the facility.F-202604-065343

Timeline	Remediation Plan
Medium Term CAP-0077488	Conduct a thorough inspection of all emergency lighting units throughout the facility to identify specific locations and units that are non-functional. This involves checking both the lights and the power sources, including batteries and wiring connections. Repair any faulty emergency lighting units immediately. If repair is not feasible, replace the defective units with new, compliant emergency lighting fixtures to ensure they provide adequate illumination during an emergency.
Longer Term CAP-0077489	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.45 - Non functional emergency light found near stitching area finishing area ground floor



Figure 2.46 - Non functional emergency light found at warehouse area ground floor

9.2 Provided but insufficient

The section refers to the entire facility.

During the inspection the emergency lighting arrangement throughout the facility was assessed.

The following observations were made:

i. Several areas lack sufficient emergency lighting coverage. The issue was noted at stitching area, cutting area, stairs, egress routes, aisle and exits.
ii. Multiple emergency luminaires indicate the need for maintenance. The issue was noted near stitching area ground floor.
iii. External emergency lighting was not provided, preventing people from reaching a place of safety outside the facility during nighttime evacuations.
iv. The facility did not maintain inspection, testing, and maintenance records for emergency lighting and exit sign systems, which is a non-compliance with NFPA 101 requirements.

Adequate and functional emergency lighting throughout the facility is crucial for providing necessary illumination during power failures or other emergencies, ensuring the safe evacuation of workers.F-202604-065347

Timeline	Remediation Plan
Medium Term CAP-0077465	i. Prepare a detailed emergency lighting design based on a comprehensive site survey to identify areas with inadequate coverage, in accordance with the Building Code of Pakistan – Fire Safety Provisions 2016. Submit the proposed emergency lighting layout to Pakistan Accord for review and approval. Upon approval, install additional emergency luminaires in the identified areas to ensure adequate illumination for safe evacuation.
Medium Term CAP-0077466	ii. Perform maintenance on all emergency luminaires indicating issues, including cleaning, bulb replacement, and verification of electrical connections.
Medium Term CAP-0077467	iii. Install external emergency luminaires to enable safe evacuation during hours of darkness all the way to a safe place outside the building(s).

Medium Term CAP-0077468	iv. Emergency lighting shall be tested at regular intervals, with functional testing carried out quarterly and full-backup duration testing conducted annually. Records of all tests and maintenance activities shall be properly maintained and kept available for inspection, in accordance with BCP Fire Safety Provisions 2016.
Longer Term CAP-0077469	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.47 - Emergency light found non-operational near stitching area



Figure 2.48 - Emergency light were not provided at aisle stitching area first floor



Figure 2.49 - Emergency lights not provided at emergency stairs outside area



Figure 2.50 - Emergency lights not provided for boiler and RO plant shed area



Figure 2.51 - Emergency lights not provided at stairs

9.3 Independent power source

The section refers to the entire facility.

During the inspection, it was observed that the emergency lighting within the facility was not connected in a manner that would ensure automatic activation upon the loss of power to the general lighting in various spaces.

Proper circuitry is crucial to ensure that emergency lighting is automatically activated when general lighting fails. This automatic activation is essential for maintaining sufficient illumination, allowing for the safe evacuation of occupants and preventing panic during power outages. Further the emergency lights and exit signs should be reconfigured to be connected on a dedicated single circuit, which must not include any other load except for the emergency lighting system. According to the Building Code of Pakistan Fire Safety Provisions 2016, emergency lighting systems must be designed to provide immediate illumination in the event of power loss to ensure a clear and safe path for evacuation.

Ensuring that the emergency lighting is integrated with the facility's power system to activate automatically during power failures is critical for life safety. This integration helps prevent hazardous conditions that can arise in the dark, such as trips, falls, and confusion, thereby facilitating orderly and efficient evacuation during emergencies.F-202604-065346

Timeline	Remediation Plan
Medium Term CAP-0077486	Reconfigure the emergency lighting circuitry to ensure that emergency lights activate immediately upon failure of general lighting power, and provide an independent circuit for the emergency lighting system, as required by the Building Code of Pakistan Fire Safety Provisions 2016. Submit circuitry design diagrams to Pakistan Accord for review.
Longer Term CAP-0077487	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Provide independent power circuit to emergency lights. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.

9.4 General Lighting

The section refers to the entire facility.

During the inspection the general lighting throughout the facility was assessed.

Several locations were identified as lacking sufficient general lighting. The issue was noted at storage area ground floor and meeting room at building 1.

Insufficient general lighting in several locations can increase the risk of accidents and impede safe evacuation during emergencies, while controlling general lighting via breakers instead of light switches poses operational inefficiencies and safety hazards. This setup can delay emergency responses and heighten the risk of electrical faults or fires due to frequent breaker use. Addressing these issues is crucial for enhancing everyday safety and ensuring effective emergency preparedness in the facility.F-202604-065379

Timeline	Remediation Plan
Medium Term CAP-0077496	Conduct a thorough assessment of lighting throughout the facility to find areas with insufficient illumination. Install additional lighting fixtures in these spots to ensure adequate brightness. Ensure the new lighting design meets safety and efficiency guidelines to provide clear visibility for all tasks and emergency situations.



Figure 2.52 - Insufficient general lighting observed at storage area ground floor



Figure 2.53 - Insufficient general lighting observed at meeting room ground floor

10. Generator, Transformer & Sub-Station Room

10.1 Ventilation and illumination

This sections refers to the electrical switchrooms, generator room(s) and transformer room(s).

An assessment of the existing ventilation system was conducted, alongside an evaluation of the lighting setup in various parts of the room, to determine their compliance with recommended standards.

The following observations were made:

i. The ventilation system was inadequate, with insufficient airflow leading to heat accumulation, particularly at LT room.
ii. Several sections of the LT room were poorly lit, with light levels falling below the minimum required for safe operation and maintenance.

Addressing these ventilation and illumination issues is crucial for life and fire safety. Proper ventilation is essential to prevent the overheating of electrical equipment, which can lead to equipment failure or fires, posing significant risks to both personnel and infrastructure. Additionally, adequate ventilation ensures that harmful fumes or gases do not accumulate, which could be hazardous to anyone entering the room. Proper illumination is equally important as it ensures that all areas are visible, reducing the risk of accidents and enabling quick identification and response to any issues that may arise.

BS7671 18th Edition + A2 (2022) specifies the need for ventilation to prevent overheating and ensure the safe operation of electrical installations. The standard also emphasises that adequate lighting must be provided to ensure visibility for safe operation and maintenance. Proper lighting levels are crucial to avoid accidents and to ensure that any issues with the electrical installation can be promptly identified and addressed. F-202604-065380

Timeline	Remediation Plan
Medium Term CAP-0077360	i. Design the ventilation layout to enhance the distribution of airflow, ensuring that critical areas around the generator and transformer receive adequate ventilation. This might involve repositioning vents or adding ductwork.
Medium Term CAP-0077361	ii. Add more lighting fixtures in poorly lit areas to ensure even distribution of light throughout the room. Focus on areas that require the highest visibility, such as around equipment and control panels.

Longer Term CAP-0077362	Establish regular maintenance and testing of the ventilation system and luminaires within the spaces to ensure adequate performance.



Figure 2.54 - Inadequate ventilation and illumination observed in LT room

10.2 Battery

During the inspection of the battery system, the battery cells, connections, and enclosures were thoroughly examined.

The following observations were noted:

i. The UPS batteries were retained inside their original cardboard packaging boxes with the top covers opened, while the battery terminals remained connected and energized. The issue was noted at 75kVA diesel generator batteries and UPS batteries at merchandiser office.
The storage of combustible cardboard material around live batteries presents a potential fire hazard. In the event of overheating, arcing, short circuit, or battery failure, the combustible packaging may contribute to fire propagation.

ii. Several battery terminals were found exposed and not properly insulated, increasing the risk of electrical shock or short circuits. The issue was noted at 75kVA diesel generator batteries and UPS batteries at merchandiser office.

Proper monthly and annual maintenance of the battery system is crucial for fire safety. Spilled fluid can cause corrosion and fires, while bare terminals increase the risk of electrical shocks and short circuits, potentially igniting fires or explosions. Non-compliant battery placement can obstruct ventilation, leading to hazardous gas buildup and fire risks. Addressing these issues is essential to prevent fires and ensure battery system safety.F-202604-065382

Timeline	Remediation Plan
Immediate CAP-0077371	i. It is recommended to remove all combustible packaging materials from around energized batteries and ensure the batteries are installed on suitable non-combustible supports/racks in accordance with good electrical safety and housekeeping practices.
Medium Term CAP-0077372	ii. Insulate exposed battery terminals using appropriate insulating materials to prevent electrical shock and short circuits. Conduct a thorough inspection of all connections to ensure that there are no immediate risks of exposure.
Longer Term CAP-0080320	Establish a routine inspection and maintenance protocol to regularly check and maintain proper insulation of all battery terminals.



Figure 2.55 - Exposed battery terminals with cardboard packing observed at diesel generator battery set in the mechanical workshop area



Figure 2.56 - Exposed battery terminals with cardboard packing observed observed in the Merchandiser office

10.3 Back-up power supply

During the assessment of the backup power supply system, an inspection was conducted on the generators, battery backups, and auxiliary power sources.

The 75kVA diesel generator exhibited signs of inadequate maintenance, corrosion, including worn components and oil accumulation inside canopy.

Maintaining a reliable backup power supply system is crucial for ensuring continuous operations and safety during power outages or emergencies. Inadequate maintenance of generators can lead to failures during critical moments, jeopardizing essential services and potentially causing disruptions or safety hazards.F-202604-065381

Timeline	Remediation Plan
Medium Term CAP-0077366	It is recommended to carry out maintenance of the diesel generator, including replacement of worn components, cleaning of oil accumulation within the canopy, and addressing areas affected by corrosion. A detailed inspection should also be conducted to identify and rectify any other issues that may impact performance.
Longer Term CAP-0077367	Establish a regular maintenance schedule for the diesel generator to ensure routine servicing, including timely replacement of worn components, cleaning of oil accumulation within the canopy, and addressing corrosion. Periodic inspections should also be carried out to identify and rectify any issues that may affect performance.



Figure 2.57 - Oil spills at 75kVA diesel generator



Figure 2.58 - Corrosion observed at diesel generator and the canopy

11. Cabling & Wiring

11.1 Cable support and protection

This section refers to the entire facility.

During the inspection of the entire facility, a visual survey of the cable containment system was conducted.

The following observations were noted:

i. Proper supports were not provided for cables; they were laid directly on the ground and hung along the walls in a PVC flexible pipe. Additionally, several locations had open and exposed electrical connections. Inadequate cable support results in sagging or stress on cables, which can lead to damage or failure. The issue was noted at boiler area, mechanical workshop area, generator area, and warehouse area.
ii. Improper or inadequate cable tray support was found at stitching area. Weak or unstable tray installations compromise the integrity of the containment system, increasing the likelihood of sagging or cable failure.
iii. Loose cables were observed inside distribution boards which were not properly terminated and left unconnected within the enclosure. This condition indicates poor workmanship and increases the risk of short circuits, arcing, and accidental energization during maintenance activities, posing a significant electrical and fire hazard. The issue was note at LT panel, stitching hall DB, stitching hall fan DB, finishing area cable tray and finishing area DB.F-202604-065335

Timeline	Remediation Plan
Medium Term CAP-0077416	i. Secure cables with appropriate supports and insulate any exposed connections to prevent sagging, stress, and potential hazards. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 522.8.5 for mechanical support and 526.5 for proper terminations)
Medium Term CAP-0077418	ii. Ensure all cable trays are properly supported and structurally sound. Replace or reinforce any weak, unstable, or inadequately mounted tray systems to maintain cable integrity and system safety. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 522.8.5).
Medium Term CAP-0077429	iii. Ensure all cables inside distribution boards are properly terminated, insulated, and securely fixed using appropriate glands and terminals in accordance with BS 7671 18th Edition + A2 (2022). Any unused or spare conductors shall be safely isolated, identified, and properly enclosed to prevent accidental contact or energization.

Longer Term CAP-0077417	i. Implement a comprehensive cable management system with regular inspections to ensure cables remain properly supported and insulated over time.
Longer Term CAP-0077419	ii. Establish a routine maintenance schedule to assess the condition and support of cable trays. Ensure proper reinforcement and alignment to prevent future sagging or instability. Document findings and corrective actions in maintenance logs.
Longer Term CAP-0077430	iii. Establish a system to ensure all DB cables are properly terminated, insulated, and secured. All spare or unused conductors shall be safely isolated, identified, and enclosed to prevent accidental contact or energization.



Figure 2.59 - Inadequate cable tray support observed at cutting hall first floor



Figure 2.60 - Loose cable found at finishing area distribution board ground floor



Figure 2.61 - Loose cables found inside LT panel at electrical room



Figure 2.62 - Inadequate cable support and protection for cables at Boiler area



Figure 2.63 - Inadequate cable support for cables near mechanical workshop area

11.2 Splicing of cables

This relates to the whole facility.

During the inspection the cable installation was visually surveyed throughout the facility.

The following observations were noted:

i. Cable terminations within panels and switchgear were made without cable lugs, and in some cases, cable strands were found cut to fit into undersized cable lugs, resulting in broken strands and loose connections. This presents a significant fire risk due to the potential for electrical arcing and overheating, which can ignite nearby materials and cause a fire. Furthermore, loose connections can cause voltage drops and equipment malfunctions, posing a life safety hazard. The issue was noted at the LT panel, stitching hall, mechanical workshop, cutting area, and stitching hall DB.
ii. PVC tape joints were found in several locations - including within distribution boards - where cables were either damaged or two cables were being joined together. This method of jointing is unreliable and can deteriorate over time, increasing the risk of electrical shorts and fires. The adhesive on PVC tape can degrade, leading to exposed conductors that can spark and ignite surrounding materials, presenting both fire and life safety risks. The issue was noted at stitching hall DB, cable tray at cutting hall, cutting hall DB, finishing area cable tray and LT panel.
iii. Improper connection of machine power plugs observed, where industrial connectors are not properly mated and are left under strain. This condition may result in loose contact, overheating, electric shock, and potential short circuit, significantly increasing fire risk and compromising personnel safety. The issue was noted at finishing area ground floor.
iv. PVC connectors used in distribution board which could be hazard of potential short circuit due to its poor insulation. The issue was noted at LT panel, and cutting area cable tray.

For any future electrical work, the termination of stranded cables must be done with cable lugs and PVC tape joints shall not be utilised. Proper termination ensures secure and reliable connections, reducing the risk of electrical failures and enhancing overall safety. F-202604-065333

Timeline	Remediation Plan
Medium Term CAP-0077454	i. Install appropriate cable lugs on all improperly terminated cables within panels and switchgear. Replace undersized lugs and ensure no cable strands are cut during termination. All connections shall be properly secured to prevent loose connections and carried out in accordance with Regulation 526.1 of BS 7671 (18th Edition + A2:2022).

Medium Term CAP-0077455	ii. Replace all PVC tape joints with proper terminal blocks or connectors to ensure secure and stable connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022). Replace all damaged cabling with new wiring.
Medium Term CAP-0077456	iii. Replace and standardize all such connections with proper compatible industrial plug and socket arrangements. Ensure all connections are fully and securely engaged with proper locking and strain relief to eliminate loose or exposed connections.
Medium Term CAP-0077457	iv. Replace all PVC connectors used inside distribution boards with approved, suitably rated insulated terminals to prevent insulation failure and potential short circuits, in accordance with Regulation 526.1 of BS 7671 (18th Edition + A2:2022).
Longer Term CAP-0077458	i. Implement a standard operating procedure (SOP) requiring the use of cable lugs for all future cable terminations. Regularly train and audit personnel to ensure compliance with this procedure, thus maintaining safe and reliable electrical connections.
Longer Term CAP-0077459	ii. Establish a policy prohibiting the use of PVC tape for electrical jointing. Regularly inspect and maintain electrical installations to ensure compliance, and provide continuous training for staff on proper jointing techniques.
Longer Term CAP-0077460	iii. Develop and enforce a facility-wide electrical connection standard requiring the exclusive use of approved industrial plug and socket systems. Conduct regular inspections and staff awareness sessions to ensure correct connection practices, proper locking, and compliance with BS 7671 requirements.
Longer Term CAP-0077461	iv. Ensure complete replacement of all PVC connectors with compliant terminations as per BS 7671 / NFPA 70 and train staff on approved practices.



Figure 2.64 - Cable termination without the use of lugs at finishing area ground floor



Figure 2.65 - PVC connector found used at the LT panel electrical room



Figure 2.66 - PVC tape joint found at cable tray installed at first floor finishing area



Figure 2.67 - Improper and unsafe power plug connections observed at finishing area ground floor

11.3 Colour coding

This relates to the whole facility.

During the inspection of several distribution boards, deviations from standard cable colour coding practices were identified. Such deviations pose significant risks, as they can lead to the misinterpretation and misidentification of cables. This misidentification can result in incorrect connections, potentially causing unsafe working conditions and increasing the likelihood of electrical accidents, such as shocks or fires. The issue was noted at cutting area DB, LT panel and first floor stitching area.

To mitigate these risks, it is imperative that all future electrical work strictly adheres to standardized colour coding as prescribed by BS 7671. This will ensure accurate identification of conductors, thereby enhancing overall safety and operational efficiency. F-202604-065336

Timeline	Remediation Plan
Medium Term CAP-0077411	Conduct a comprehensive audit of all distribution boards to identify and correct any deviations from the standard cable colour coding. This involves re-labelling or re-insulating the cables with the appropriate colour codes as specified in BS 7671. Specifically, adhere to BS 7671 Regulation 514.3.1, which mandates the correct identification of conductors.
Longer Term CAP-0077412	Develop and enforce a strict protocol for cable colour coding to be followed in all future installations and maintenance activities. This protocol should include regular training sessions for electricians and maintenance personnel to keep them updated on the latest standards and practices. Additionally, periodic audits should be conducted to ensure ongoing compliance with the standardized colour coding system, thereby preventing future deviations and ensuring sustained safety and operational efficiency.



Figure 2.68 - Color coding violation observed at cutting area DB



Figure 2.69 - Color coding violation observed at LT panel electrical room



Figure 2.70 - Color coding violation observed at first floor stitching area DB

11.4 Fixtures

This relates to the whole facility.

During the inspection, the electrical fixtures throughout the facility were visually surveyed.

The following observations were noted:

i. Power receptacles mounted on combustible wooden ply: Installing receptacles on combustible materials like wooden ply increases the risk of fire if a short circuit or overheating occurs. Combustible materials can easily ignite, leading to potential fire hazards and endangering life safety. The issue was noted at HR and admin office, Lab room, accessory store, outside main building and mechanical workshop.
ii. LED lights were found suspended directly from cables, and several fixtures were broken due to lack of proper support, particularly in the electronics workshop and stitching hall on the second floor of the main production building, as well as boiler control room. The issue was noted at boiler area and near stitching area.
iii. Burnt and heat-damaged power sockets: Several power sockets in the packing area were found with visible signs of burning and discoloration. This condition indicates overheating or arcing, which can lead to equipment failure or ignition of surrounding materials. Such deterioration significantly increases the risk of electrical fires and poses a serious life safety hazard. F-202604-065329

Timeline	Remediation Plan
Medium Term CAP-0077383	i. Replace combustible wooden ply backings with non-combustible materials to reduce fire risk.
Medium Term CAP-0077384	ii. Install appropriate lighting fixtures with secure mounting provisions to support bulbs and LED tube lights independently, without relying on electrical cables.
Medium Term CAP-0077392	iii. Replace all burnt or heat-damaged power sockets with new, properly rated receptacles and ensure secure electrical connections.
Longer Term CAP-0077385	i. Establish a facility-wide standard to use only non-combustible materials for mounting electrical components, and conduct regular audits for compliance.

Longer Term CAP-0077386	ii. Establish and implement a facility-wide standard for the proper installation of all lighting fixtures using durable supports, to prevent stress on electrical wiring and ensure long-term safety and reliability.
Longer Term CAP-0077393	iii. Establish a preventive maintenance and periodic inspection program for all electrical sockets to identify and address early signs of damage or overheating.



Figure 2.71 - Wooden switchboard found at accessory store ground floor



Figure 2.73 - LED fixture improperly suspended by its cable near stitching area



Figure 2.72 - Wooden switchboard found at at HR and Admin office



Figure 2.74 - Sign of burning on a switchboard found at sample room

12. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.